MA 17: HOW TO SOLVE IT
HANDOUT 4: ENUMERATIVE COMBINATORICS, PROBABILITY

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Stars and bars. The equation $x_1 + \cdots + x_r = n$ has $\binom{n+r-1}{r-1}$ solutions in nonnegative integers: consider n stars and $r-1$ bars in a row.

Problem 1.
Let n, k be positive integers with $n \geq 2k-1$. Prove that the number of k-element subsets of $\{1, 2, \ldots, n\}$ containing no two consecutive integers is

$$\binom{n-k+1}{k}.$$

Hint: The selected integers divide the set into $k+1$ 'gaps'.

Problem 2.
For a positive integer n, count the sequences $(a_1, \ldots, a_{2n})$ with each $a_i \in \{-1, 1\}$ such that

$$\sum_{i=1}^{2n} a_i = 0, \qquad \sum_{i=1}^{j} a_i \geq 0 \quad (1 \leq j \leq 2n).$$

Hint: In a 'bad' sequence, reverse every sign up to the first time its partial sum is -1.

Generating functions. For a sequence $a_0, a_1, \ldots$, its generating function is $F(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$, which can be used for combinatorial enumeration and solving recurrence relations. (For a finite sequence, set the remaining coefficients to zero.)

Problem 3. (Putnam 2020, A2)
Let k be a nonnegative integer. Evaluate

$$\sum_{j=0}^{k} 2^{k-j} \binom{k+j}{j}.$$

Hint: Find a function where this sum is the k-th coefficient.

Probability.
Sample space: the set of possible outcomes. An event is a subset of the sample space, and probabilities are assigned to events.
Example 1: Choose a card uniformly from 54 cards containing four kings. Each card has probability $\frac{1}{54}$, and a king is chosen with probability $\frac{4}{54} = \frac{2}{27}$.
Example 2: If X is uniform on $[0, 1]$, then $\mathbb{P}(0.3 \leq X \leq 0.9) = 0.6$, while each individual point has probability zero.

Problem 4. (Canadian Mathematical Olympiad 1980, Problem 4)
A gambling student tosses a fair coin and scores one point for each head that turns up and two points for each tail. Prove that the probability of the student scoring *exactly n points* is $\frac{1}{3}[2+(-\frac{1}{2})^n]$.

Hint: Condition on the first toss.

Problem 5. (Putnam 2021, B1)
Suppose that the plane is tiled with an infinite checkerboard of unit squares. If another unit square is dropped on the plane at random with position and orientation independent of the checkerboard tiling, what is the probability that it does not cover any of the corners of the squares of the checkerboard?

Hint: Fix the orientation and place the square's center uniformly in one unit cell.

Expectation.
For a nonnegative integer-valued random variable X, its expectation is

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k\mathbb{P}(X = k).$$

If X has density f, then $\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x)\,dx$ when this integral is absolutely convergent. Expectation is linear: for random variables X and Y with finite expectations, $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$; independence is not required.

Problem 6.
Choose $2n$ points $A_1, B_1, \ldots, A_n, B_n$ independently and uniformly on a circle, and draw the chords A_iB_i. Find the expected number of intersection points in the open disk and the expected number of regions into which the chords divide the disk.

Hint: Consider each pair of crossing chords and the cyclic order of their endpoints. For the number of regions, add the chords one at a time.

Problem 7. (Putnam 2022, A4)
Suppose that $X_1, X_2, \ldots$ are real numbers between 0 and 1 that are chosen independently and uniformly at random. Let $S = \sum_{i=1}^{k} X_i/2^i$, where k is the least positive integer such that $X_k < X_{k+1}$, or $k = \infty$ if there is no such integer. Find the expected value of S.

Hint: Term i is included exactly when $X_1 \geq X_2 \geq \cdots \geq X_i$. Compute its expected contribution by conditioning on $X_i = t$.

Problem 8.
Alice starts with m coins and Bob with n coins, where m, n are positive integers. Each round, an independent fair coin toss decides who receives one coin from the other player. They stop when one player has no coins. Find the expected number of rounds.

Hint: Keep $N = m + n$ fixed. Let e_i be the expected remaining time when Alice has i coins. Write the boundary conditions and the first-step equation.

Problem 9. (Putnam 2020, B3)
Let $x_0 = 1$, and let δ be some constant satisfying $0 < \delta < 1$. Iteratively, for $n = 0, 1, 2, \ldots$, a point x_{n+1} is chosen uniformly from the interval $[0, x_n]$. Let Z be the smallest value of n for which $x_n < \delta$. Find the expected value of Z, as a function of δ.

Hint: Let $e(x)$ be the expected remaining number of steps from x. Condition on the next choice.

Reminder PSet 4 to be released tomorrow, and due by the end of Nov 11 (on Canvas).